

LINUX FILESYSTEM

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Linux Filesystem Investigation Assignment

Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

Determine:

- The storage devices available.
- The partitions available on each device.
- The filesystem type used by each partition.
- The device/partition on which `/` is located.
- The filesystem type used for `/`.
- The total, used, and available storage space.

Suggested commands

```
lsblk
lsblk -f
df -T
df -h
```

Answers

Question	Command	Output / Observation	Explanation / Answer
1. What storage devices are available?	lsblk	sda — 20G — disk; sr0 — 1024M — rom	Observe `NAME`, `SIZE`, and `TYPE`. The main storage device is `/dev/sda` (20G).
2. What partitions are available on each device?	lsblk	sda1 — 18.9G — part; sda2 — 1K — part; sda5 — 1.1G — part	Entries with `TYPE = part` under `sda` are the partitions of `/dev/sda`.
3. What filesystem type is used by each partition?	lsblk -f	/dev/sda1 → ext4; /dev/sda5 → swap; /dev/sda2 → no filesystem	The `FSTYPE` column shows the filesystem type.
4. On which device/partition is `/` located?	df -T	/dev/sda1 ext4 ... /	The row containing `/` identifies the root partition. Answer `/dev/sda1`.
5. What filesystem type is used for `/`?	df -T	/dev/sda1 ext4 ... /	The `Type` column shows `ext4`.
6. What is the total, used, and available storage space?	df -h	/dev/sda1 19G 15G 2.7G 85% /	Total = 19G, Used = 15G, Available = 2.7G.

Storage Configuration Summary

Device/Partition	Filesystem Type	Size	Mount Point
/dev/sda	—	20G	—
/dev/sda1	ext4	18.9G	/
/dev/sda2	—	1K	—
/dev/sda5	swap	1.1G	[SWAP]
/dev/sr0	—	1024M	—

```

(gopika@kali)-[~/linux/theory/t3]
$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   20G  0 disk
├─sda1       8:1    0  18.9G  0 part /
├─sda2       8:2    0    1K  0 part
├─sda5       8:5    0   1.1G  0 part [SWAP]
└─sr0       11:0    1 1024M  0 rom

(gopika@kali)-[~/linux/theory/t3]
$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1      ext4    1.0             fc73c44a-d5bb-4835-bf68-b9f53b23571f   2.6G    80% /
├─sda2
├─sda5      swap    1              2d3ca6d1-485a-434c-88c4-cc10d93fb6c6      [SWAP]
└─sr0

(gopika@kali)-[~/linux/theory/t3]
$ df -T
Filesystem      Type      1K-blocks      Used Available Use% Mounted on
udev            devtmpfs   972828            0    972828    0% /dev
tmpfs           tmpfs      201664           980    200684    1% /run
/dev/sda1       ext4      19353424 15578256 2766736   85% /
tmpfs           tmpfs      1008300            4    1008296    1% /dev/shm
tmpfs           tmpfs      1008304            8    1008296    1% /tmp
none            tmpfs      1024              0      1024    0% /run/credentials/getty@tty1.service
tmpfs           tmpfs      201660          108    201552    1% /run/user/1000
none            tmpfs      1024              0      1024    0% /run/credentials/systemd-journald.service

(gopika@kali)-[~/linux/theory/t3]
$ df -h
Filesystem      Size      Used Avail Use% Mounted on
udev            951M        0    951M   0% /dev
tmpfs           197M      980K    196M   1% /run
/dev/sda1       19G       15G   2.7G  85% /
tmpfs           985M     4.0K    985M   1% /dev/shm
tmpfs           985M     8.0K    985M   1% /tmp
none            1.0M        0     1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M     108K    197M   1% /run/user/1000
none            1.0M        0     1.0M   0% /run/credentials/systemd-journald.service

(gopika@kali)-[~/linux/theory/t3]
$ df -T /
Filesystem      Type 1K-blocks      Used Available Use% Mounted on
/dev/sda1       ext4 19353424 15578260 2766732   85% /

(gopika@kali)-[~/linux/theory/t3]
$ df -h /
Filesystem      Size      Used Avail Use% Mounted on
/dev/sda1       19G       15G   2.7G  85% /

```

Exercise 2 – Disk → Partition → Filesystem → Mount Point

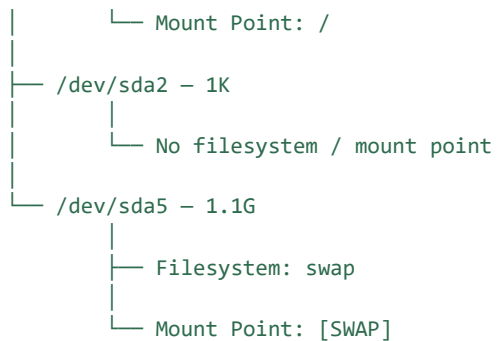
Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.

Based on the values observed from your Kali Linux system:

```

/dev/sda – 20G Disk
|
├─ /dev/sda1 – 18.9G
|   |
|   └─ Filesystem: ext4
|
└─

```



`/dev/sr0` is a separate **1024 MB ROM/CD/DVD device**, so it is not part of the /dev/sda` disk hierarchy.`

Explain in 2–3 sentences the relationship between these four components.

A disk is the main storage device, which is divided into partitions. Each partition can contain a filesystem, and the filesystem is attached to a mount point so that Linux can access and organize the stored files.

For your system, you can write:

`/dev/sda` is the 20G disk that contains partitions such as /dev/sda1` and /dev/sda5`. /dev/sda1` uses the ext4` filesystem and is mounted at /`, which is the main root filesystem of your Kali Linux system.`

```
Session Actions Edit View Help

(gopika@kali)-[~/linux/theory/t3]
$ pwd
/home/gopika/linux/theory/t3

(gopika@kali)-[~/linux/theory/t3]
$ cd /

(gopika@kali)-[/]
$ ls /
bin  dev  home  initrd.img.old  lib32  lost+found  mnt  proc  run  srv  tmp  var  vmlinuz.old
boot  etc  initrd.img  lib  lib64  media  opt  root  sbin  sys  usr  vmlinuz

(gopika@kali)-[/]
$ ls -d */
/bin/  /dev/  /home/  /lib32/  /lost+found/  /mnt/  /proc/  /run/  /srv/  /tmp/  /var/
/boot/  /etc/  /lib/  /lib64/  /media/  /opt/  /root/  /sbin/  /sys/  /usr/

(gopika@kali)-[/]
$ ls -la /
total 76
drwxr-xr-x 19 root root 4096 Aug 11 01:22 .
drwxr-xr-x 19 root root 4096 Aug 11 01:22 ..
lrwxrwxrwx 1 root root 7 Aug 11 00:47 bin -> usr/bin
drwxr-xr-x 3 root root 4096 Aug 11 01:32 boot
drwx----- 2 root root 4096 Aug 11 01:22 .cache
drwxr-xr-x 18 root root 3280 Sep 3 09:29 dev
drwxr-xr-x 184 root root 12288 Sep 3 09:29 etc
drwxr-xr-x 6 root root 4096 Sep 1 18:23 home
lrwxrwxrwx 1 root root 34 Aug 11 00:50 initrd.img -> boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx 1 root root 34 Aug 11 00:50 initrd.img.old -> boot/initrd.img-6.19.14+kali-amd64
lrwxrwxrwx 1 root root 7 Aug 11 00:47 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Aug 11 01:07 lib32 -> usr/lib32
lrwxrwxrwx 1 root root 9 Aug 11 00:47 lib64 -> usr/lib64
drwx----- 2 root root 16384 Aug 11 00:47 lost+found
drwxr-xr-x 3 root root 4096 Aug 11 00:47 media
drwxr-xr-x 2 root root 4096 Aug 11 00:47 mnt
drwxr-xr-x 3 root root 4096 Aug 11 01:06 opt
dr-xr-xr-x 229 root root 0 Sep 3 09:29 proc
drwx----- 4 root root 4096 Sep 3 09:29 root
drwxr-xr-x 41 root root 960 Sep 3 09:30 run
lrwxrwxrwx 1 root root 8 Aug 11 00:47 sbin -> usr/sbin
drwxr-xr-x 3 root root 4096 Aug 11 01:11 srv
dr-xr-xr-x 13 root root 0 Sep 3 09:29 sys
drwxrwxrwt 11 root root 300 Sep 3 18:20 tmp
drwxr-xr-x 15 root root 4096 Aug 11 01:07 usr
drwxr-xr-x 12 root root 4096 Aug 11 01:16 var
lrwxrwxrwx 1 root root 31 Aug 11 00:50 vmlinuz -> boot/vmlinuz-6.19.14+kali-amd64
lrwxrwxrwx 1 root root 31 Aug 11 00:50 vmlinuz.old -> boot/vmlinuz-6.19.14+kali-amd64
```

Exercise 3 – Explore the Root Filesystem

Start from your current location.

Question	Command	Output / Observation	Explanation
1. Determine your current working directory.	pwd	/home/gopika/linux/theory/t3	Shows the directory where you are currently working.
2. Navigate to the root directory.	cd / pwd	No output from `cd`; `pwd` should show `/`.	Confirms that you are in the root directory.
3. Display the contents of the root directory.	ls /	bin boot dev etc home lib media mnt opt proc root run sbin srv sys tmp usr var	Shows names directly under `/`.
4. Identify all directories directly available under `/`.	ls -d */	/bin/ /boot/ /dev/ /etc/ /home/ /lib/ /media/ /mnt/ /opt/ /proc/ /root/ /run/ /sbin/ /srv/ /sys/ /tmp/ /usr/ /var/	Entries ending with `/` are directories directly under `/`.
5. Determine whether hidden entries are present.	ls -la /	Listing includes `.` and `..`, and any other entries beginning with `.` if present.	`-a` displays hidden entries.

Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations: `/home` `/root` `/etc` `/var` `/var/log` `/tmp` `/usr` `/boot` `/dev` `/proc`

For each location:

- Examine its contents.
- Identify at least three entries, where available.
- Determine the likely purpose of the directory.

Directory	Operation	Observed Contents	Purpose
<code>/home</code>	<code>ls /home</code>	<code>gopika</code>	Stores personal files and directories of normal users.
<code>/root</code>	<code>sudo ls -l /root</code>	<code>total 0 — empty</code>	Home directory of the root user.
<code>/etc</code>	<code>ls /etc</code>	<code>apt, ssh, systemd</code>	Contains system-wide configuration files and directories.
<code>/var</code>	<code>ls /var</code>	<code>cache, lib, log</code>	Stores variable data such as logs, caches, and application data.
<code>/var/log</code>	<code>ls /var/log</code>	<code>auth.log, syslog, journal</code>	Stores system and application log information.
<code>/tmp</code>	<code>ls /tmp</code>	Temporary files and directories	Stores temporary data created by users and applications.
<code>/usr</code>	<code>ls /usr</code>	<code>bin, lib, share</code>	Contains user programs, libraries, and shared resources.
<code>/boot</code>	<code>ls /boot</code>	<code>grub, vmlinuz, initrd.img</code>	Contains files required to boot the Linux system.
<code>/dev</code>	<code>ls /dev</code>	<code>null, random, tty</code>	Contains special device files used to access hardware and virtual devices.
<code>/proc</code>	<code>ls /proc</code>	<code>1, cpuinfo, meminfo</code>	Virtual filesystem providing information about running processes and the Linux kernel.

```
(gopika@kali)-[~/linux/theory/t3]
$ ls /home
alice bab charlie gopika

(gopika@kali)-[~/linux/theory/t3]
$ ls /root
ls: cannot open directory '/root': Permission denied

(gopika@kali)-[~/linux/theory/t3]
$ sudo ls /root

(gopika@kali)-[~/linux/theory/t3]
$ ls -l /root
ls: cannot open directory '/root': Permission denied

(gopika@kali)-[~/linux/theory/t3]
$ sudo ls -l /root
total 0

(gopika@kali)-[~/linux/theory/t3]
$ ls /etc
adduser.conf      firefox-esr      libccid_Info.plist  passwd          speech-dispatcher
alternatives      fonts           libnl-3             passwd-         sqlmap
apache2           freetds         libpaper.d          perl            ssh
apparmor          fstab           lightdm             nhp             ssl
```

```

(gopika@kali)-[~/linux/theory/t3]
$ ls /var
backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp  www

(gopika@kali)-[~/linux/theory/t3]
$ ls /var/log
alternatives.log  boot.log.2  inetsim  macchanger.log  openvpn  samba  wtmp.db
apache2          btmp        installer  mosquito        postgresql  speech-dispatcher  Xorg.0.log
apt              dpkg.log    journal    mysql            private    stunnel           Xorg.0.log.old
boot.log         fontconfig.log  lastlog    nginx            redis      sysstat
boot.log.1      gvm         lightdm    notus-scanner    runit      wtmp

(gopika@kali)-[~/linux/theory/t3]
$ ls /tmp
config-err-F4elbm
systemd-private-fde8891f4a6247e3b1d0d2ad4cd1f755-color.service-I1dYgy
systemd-private-fde8891f4a6247e3b1d0d2ad4cd1f755-ModemManager.service-Q6RQIC
systemd-private-fde8891f4a6247e3b1d0d2ad4cd1f755-polkit.service-8dKdNI
systemd-private-fde8891f4a6247e3b1d0d2ad4cd1f755-systemd-logind.service-pvQ194
systemd-private-fde8891f4a6247e3b1d0d2ad4cd1f755-upower.service-oqCz8H

(gopika@kali)-[~/linux/theory/t3]
$ ls /usr
bin  games  i686-w64-mingw32  include  lib  lib32  lib64  libexec  local  sbin  share  src  x86_64-w64-mingw32

(gopika@kali)-[~/linux/theory/t3]
$

```

Exercise 5 – Find Your Home Directory

Without assuming its location:

Question	Command	Output / Observation	Explanation
1. Determine your username.	whoami	gopika	Displays the username of the currently logged-in user.
2. Determine your home directory.	echo \$HOME	/home/gopika	Displays the home directory assigned to the current user.
3. Navigate to your home directory.	cd "\$HOME"	No output	Moves to the user's home directory without manually specifying its path.
4. Display its absolute path.	pwd	/home/gopika	Displays the complete path of the current working directory.
5. List its normal contents.	ls	Your actual files and directories	Displays normal, non-hidden contents.
6. List all contents, including hidden files.	ls -la	Your files, directories, `.` , `..` , and hidden entries	`-a` displays hidden entries; `-l` provides detailed information.

Exercise 6 – Hidden Files Investigation

Question	Command	Output / Observation	Explanation
Compare normal contents	ls	Desktop, Documents, Downloads, Music, Pictures, Projects, Videos	Normal `ls` does not display hidden entries beginning with `.`.
Compare all contents	ls -a	Normal entries along with `.` , `..` , `.` , `..` , `.` , `..` , `.` , `..` , etc.	`-a` displays all entries, including hidden files and directories.
Identify at least 2 hidden files/directories	ls -a	.` , `..` , `.` , `..` , `.` , `..` , `.` , `..` , etc.	Hidden because their names begin with `.`.

Answers

1. How does Linux identify a hidden file?

A file or directory is hidden when its name begins with `.`.

2. Why does a normal `ls` not display these files?

`ls` hides entries beginning with `.` by default.

3. What do `.` and `..` mean?

`.` means the current directory, while `..` means the parent directory.

4. Can a hidden file still be accessed if its name is known?

Yes. For example, `.`.bashrc` can be accessed using `cat ~/.bashrc`.

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

Hidden files can be legitimate configuration files such as `.`.bashrc` and `.`.zshrc`. However, malicious files can also be hidden, so their name alone does not determine whether they are safe or harmful.

```
(gopika@kali)-[~]
$ ls
Desktop  Downloads  lab.sh.save  linuxxhw  nmap1  Projects  Templates
Documents  ethical  linux  Music  Pictures  Public  Videos

(gopika@kali)-[~]
$ ls -a
.          .java          .vboxclient-clipboard-tty7-servic
..         .john          .vboxclient-draganddrop-tty7-cont
.bash_logout  lab.sh.save    .vboxclient-draganddrop-tty7-serv
.bashrc       linux          .vboxclient-hostversion-tty7-cont
.bashrc.original  linuxxhw      .vboxclient-seamless-tty7-control
.cache        .local         .vboxclient-seamless-tty7-service
.config       .mozilla       .vboxclient-vmsvga-session-tty7-d
Desktop       Music          .vboxclient-vmsvga-session-tty7-s
Documents     Pictures
Downloads     .profile
ethical       Projects
.face         Public
.face.icon    .ssh
.gitconfig    .sudo_as_admin_successful
.gnupg        Templates
.ICEauthority .vboxclient-clipboard-tty7-control.pid
```

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: `mkdir`

Hint: you can create this hierarchy in a single command

```
cyber_lab/
|
+-- evidence/
|   +-- network/
|   +-- system/
|
+-- logs/
|
+-- scripts/
|
+-- reports/
```

Create the following files:

```
cyber_lab/evidence/network/traffic.txt
cyber_lab/evidence/system/processes.txt
cyber_lab/logs/security.log
cyber_lab/scripts/check.sh
cyber_lab/reports/investigation.txt
```

Use appropriate Linux commands to create the structure.

Verify that the required hierarchy has been created successfully.

Hint: `tree`

Explore options

Task	Command	Output / Observation	Explanation
Create the directory hierarchy	<code>mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports</code>	No output	<code>`mkdir`</code> creates directories. <code>`-p`</code> creates required parent directories automatically.
Create the required files	<code>touch cyber_lab/evidence/network/traffic.txt cyber_lab/evidence/system/processes.txt cyber_lab/logs/security.log cyber_lab/scripts/check.sh cyber_lab/reports/investigation.txt</code>	No output	<code>`touch`</code> creates empty files with the specified names.
Verify the hierarchy	<code>tree cyber_lab</code>	Displays <code>`cyber_lab`</code> with <code>`evidence`</code> , <code>`logs`</code> , <code>`scripts`</code> , <code>`reports`</code> , and required files.	<code>`tree`</code> displays the directory structure in a tree-like format.
Explore <code>`tree`</code> options	<code>tree --help</code>	Displays available <code>`tree`</code> options.	Shows options used to control the directory display.

```
(gopika@kali)~[~/linux/theory/t3]
$ mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports

(gopika@kali)~[~/linux/theory/t3]
$ ls
cyber_lab

(gopika@kali)~[~/linux/theory/t3]
$ touch cyber_lab/evidence/network/traffic.txt cyber_lab/evidence/system/processes.txt cyber_lab/logs/security.log
cyber_lab/scripts/check.sh cyber_lab/reports/investigation.txt

(gopika@kali)~[~/linux/theory/t3]
$ tree cyber_lab
cyber_lab
├── evidence
│   ├── network
│   │   └── traffic.txt
│   └── system
│       └── processes.txt
├── logs
│   └── security.log
├── reports
│   └── investigation.txt
├── scripts
│   └── check.sh
└── 7 directories, 5 files

(gopika@kali)~[~/linux/theory/t3]
$ tree --help
usage: tree [-acdfghilnpqrstuvxACDFJQNSUX] [-L level [-R]] [-H [-]baseHREF]
          [-T title] [-o filename] [-P pattern] [-I pattern] [--gitignore]
          [--gitfile[=]file] [--matchdirs] [--metafirst] [--ignore-case]
```

Exercise 8 – Relative and Absolute Paths

Assume the current directory is:

`~/cyber_lab`

Question	Command	Output / Observation	Explanation
1. Navigate to <code>`evidence/network`</code> using a relative path.	<code>cd evidence/network</code> <code>pwd</code>	<code>~/cyber_lab/evidence/network</code>	The path starts from the current directory, so it is relative.
2. Navigate to <code>`/etc`</code> using an absolute path.	<code>cd /etc</code> <code>pwd</code>	<code>/etc</code>	The path starts from <code>`/`</code> , so it is absolute.
3. Return to your home directory using the shortest possible command.	<code>cd</code> <code>pwd</code>	<code>/home/gopika</code>	<code>`cd`</code> without a path directly takes you to your home directory.
4. Navigate to <code>`cyber_lab/reports`</code> .	<code>cd ~/cyber_lab/reports</code> <code>pwd</code>	<code>/home/gopika/cyber_lab/reports</code>	<code>`~`</code> represents your home directory.
5. Move one directory upward.	<code>cd ..</code> <code>pwd</code>	<code>/home/gopika/cyber_lab</code>	<code>`..`</code> represents the parent directory.

Question	Command	Output / Observation	Explanation
6. Display the current directory after every operation.	pwd	Displays the current absolute path	`pwd` shows the exact current location.

Definitions

Absolute path:

A path that starts from the root `/` and gives the complete location of a file or directory. Example: `/home/gopika/cyber\_lab/reports`

Relative path:

A path that starts from the current working directory instead of `/`. Example: `evidence/network`

Difference between `/` and `~`:

- `/` → Root directory, the starting point of the entire Linux filesystem.
- `~` → Home directory of the currently logged-in user.

/ → Root of Kali Linux
~ → /home/gopika

```
(gopika@kali)-[~/linux/theory/t3/cyber_lab]
$ cd evidence/network

(gopika@kali)-[~/linux/theory/t3/cyber_lab/evidence/network]
$ pwd
/home/gopika/linux/theory/t3/cyber_lab/evidence/network

(gopika@kali)-[~/linux/theory/t3/cyber_lab/evidence/network]
$ cd /etc

(gopika@kali)-[/etc]
$ pwd
/etc

(gopika@kali)-[/etc]
$ cd

(gopika@kali)-[~]
$ pwd
/home/gopika

(gopika@kali)-[~/linux/theory/t3]
$ file cyber_lab/evidence/network/traffic.txt
cyber_lab/evidence/network/traffic.txt: empty

(gopika@kali)-[~/linux/theory/t3]
$ ls -l cyber_lab/evidence/network/traffic.txt
-rw-rw-r-- 1 gopika gopika 0 Sep  3 18:52 cyber_lab/evidence/network/traffic.txt
```

Exercise 9 – Identify File Types

Investigate the following:

`/etc/passwd` `/etc` `/bin/bash` `/dev/null` `/proc`

Also investigate the files created in Exercise 7.

```
file <filename>
ls -l <filename>
```

Classify each object.

Use the two commands separately for each object:

- `file <filename>` → identifies the type of the object.
- `ls -l <filename>` → shows details and helps confirm the object type from the first character of the permissions.

Object	`file` command	`ls -l` command	Type
/etc/passwd	file /etc/passwd Output: ASCII text or similar	ls -l /etc/passwd Observe first character `-`	Regular file

Object	`file` command	`ls -l` command	Type
/etc	file /etc Output: directory	ls -ld /etc Observe first character `d`	Directory
/bin/bash	file /bin/bash Output: ELF ... executable	ls -l /bin/bash Observe first character `-`	Executable file
/dev/null	file /dev/null Output: character special	ls -l /dev/null Observe first character `c`	Character device
/proc	file /proc Output: directory	ls -ld /proc Observe first character `d`	Directory / virtual filesystem
security.log	file cyber_lab/logs/security.log Output: empty if created using touch	ls -l cyber_lab/logs/security.log Observe first character `-`	Regular file
check.sh	file cyber_lab/scripts/check.sh Output: empty if created using touch	ls -l cyber_lab/scripts/check.sh Observe first character `-`	Regular file

Final Classification

Object	Type
/etc/passwd	Regular file
/etc	Directory
/bin/bash	Executable file
/dev/null	Character device
/proc	Directory / virtual filesystem
security.log	Regular file
check.sh	Regular file

Important

For `/etc` and `/proc`, use:

```
ls -ld /etc
ls -ld /proc
```

instead of `ls -l /etc` because `ls -l /etc` shows the contents inside the directory. The `-d` option shows information about the directory itself.

```
- → Regular file
d → Directory
c → Character device
b → Block device
l → Symbolic link
```

```
-rw-r--r-- → Regular file
drwxr-xr-x → Directory
crw-rw-rw- → Character device
```

```
(gopika@kali)-[~]
$ ls -ld /etc
drwxr-xr-x 184 root root 12288 Sep  3 09:29 /etc

(gopika@kali)-[~]
$ file /etc
/etc: directory

(gopika@kali)-[~]
$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3387 Sep  1 18:23 /etc/passwd

(gopika@kali)-[~]
$
```

Exercise 10 – Examine File Metadata

Use the file:

```
~/linux/theory/t3/cyber lab/reports/investigation.txt
```

1. Add a small amount of text

```
cd ~/linux/theory/t3/cyber_lab/reports
echo "Security investigation report" > investigation.txt
```

2. Examine the file

Run these separately:

```
ls -l investigation.txt
stat investigation.txt
```

```
(gopika@kali)~[/linux/theory/t3/cyber_lab/reports]
$ echo "Security investigation report" > investigation.txt

(gopika@kali)~[/linux/theory/t3/cyber_lab/reports]
$ ls -l investigation.txt
-rw-rw-r-- 1 gopika gopika 30 Sep  3 19:18 investigation.txt

(gopika@kali)~[/linux/theory/t3/cyber_lab/reports]
$ stat investigation.txt
  File: investigation.txt
  Size: 30          Blocks: 8          IO Block: 4096   regular file
Device: 8,1        Inode: 669079       Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/  gopika)   Gid: ( 1000/  gopika)
Access: 2026-09-03 18:52:48.868541626 +0530
Modify: 2026-09-03 19:18:50.148650259 +0530
Change: 2026-09-03 19:18:50.148650259 +0530
 Birth: 2026-09-03 18:52:48.868541626 +0530

(gopika@kali)~[/linux/theory/t3/cyber_lab/reports]
$ ls -l investigation.txt /stat investigation.txt
ls: cannot access 'stat': No such file or directory
-rw-rw-r-- 1 gopika gopika 30 Sep  3 19:18 investigation.txt
-rw-rw-r-- 1 gopika gopika 30 Sep  3 19:18 investigation.txt

/:
```

3. Metadata

Item	Command	What to Observe	Answer / Meaning
Filename	stat investigation.txt	File: investigation.txt	investigation.txt
File size	stat investigation.txt	Size: 30	30 bytes
File type	stat investigation.txt	regular file	Regular file
Owner and Group	ls -l investigation.txt	gopika gopika	Owner = gopika, Group = gopika
Inode number	stat investigation.txt	Inode: <number>	Unique number identifying the file
Access time	stat investigation.txt	Access: <date/time>	Last time the file was accessed
Modification time	stat investigation.txt	Modify: <date/time>	Last time the file contents were modified
Change time	stat investigation.txt	Change: <date/time>	Last time the file's metadata/status was changed

Your observed `ls -l` output

```
-rw-rw-r-- 1 gopika gopika 30 Sep 3 19:18 investigation.txt
```

- Filename: `investigation.txt`
- Size: `30 bytes`
- Owner: `gopika`
- Group: `gopika`
- Type: Regular file

For the remaining values, use the actual output of:

stat investigation.txt

Answers

1. What is metadata?

Metadata is information about a file, such as its name, size, owner, permissions, inode number, and timestamps.

2. Is metadata the same as file content?

No. File content is the actual data stored in the file, while metadata describes the file.

3. Which timestamp changes when file contents are modified?

The `Modify` timestamp changes when the contents are modified. The `Change` timestamp normally changes as well because the file's status is updated.

Exercise 11 – Explore `/etc`

These commands only **read/display** the files; they do not modify them.

Investigation

Item	Command	What to Observe	Purpose / Information
/etc	ls /etc	passwd, group, hostname, hosts and many other configuration files	Contains system-wide configuration files and directories.
passwd	cat /etc/passwd	Lines containing usernames and account information	Stores user account information such as username, UID, GID, home directory, and login shell.
group	cat /etc/group	Group names and associated group IDs/users	Stores information about groups configured on the system.
hostname	cat /etc/hostname	Your Kali system's hostname	Contains the hostname assigned to the system.
hosts	cat /etc/hosts	IP addresses mapped to hostnames	Provides local hostname-to-IP address mappings.

Answers

1. What kind of information is primarily stored under `/etc`?

`/etc` primarily stores system-wide configuration files used by Linux and its applications.

2. What information does `/etc/passwd` contain?

It contains information about user accounts, including usernames, UIDs, GIDs, home directories, and login shells. It does not normally contain the actual user passwords.

3. What information does `/etc/hostname` contain?

It contains the hostname of the Linux system, which is the name used to identify the system on a network.

4. Why is `/etc` security-sensitive?

`/etc` contains important system and security configuration files. Unauthorized modification could change user accounts, authentication, networking, services, or other system behavior.

```
(gopika@kali) - [~/linux/theory/t3]
$ ls /etc
adduser.conf          firefox-esr           libccid_Info.plist  passwd              speech-dispatcher
alternatives          fonts                libnl-3             passwd-            sqlmap
apache2               freetds              libpaper.d          perl               ssh
apparmor              fstab                lightdm             php                ssl
apparmor.d            fuse.conf            lighttpd            plymouth           sslsplit
apt                   gai.conf             locale.conf         polkit-1           strongswan.conf
arp-scan              geoclue              locale.gen          postgresql         strongswan.d
avahi                 ghostscript          localtime           postgresql-common stunnel
bash.bashrc           glvnd                logcheck            powershell-empire subgid
bash_completion       gnome-system-tools  login.defs          ppp                subgid
bash_completion.d     gophish              logrotate.conf     profile            subuid
bindresvport.blacklist gprofng.rc           logrotate.d         profile.d           subuid
binfmt.d              group                magic               protocols          subversion
bluetooth             group               machine-id          proxychains4.conf  sudo.conf
ca-certificates       group-              magic.mime          python2.7           sudoers
ca-certificates.conf grub.d               magic.mime          python2.7           sudoers.d
```

```

(gopika@kali)-[~/linux/theory/t3]
$ ls /etc | head -n 5
adduser.conf
alternatives
apache2
apparmor
apparmor.d

(gopika@kali)-[~/linux/theory/t3]
$ cat /etc/passwd | head -n 5
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync

(gopika@kali)-[~/linux/theory/t3]
$ cat /etc/group | head -n 5
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:gopika

(gopika@kali)-[~/linux/theory/t3]
$ cat /etc/group | head -n 5
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:gopika

(gopika@kali)-[~/linux/theory/t3]
$ ls /var/log | head -n 5
alternatives.log
apache2
apt
boot.log
boot.log.1

(gopika@kali)-[~/linux/theory/t3]
$ lsblk | head -n 5
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   20G  0 disk
├─sda1        8:1    0  18.9G  0 part /
├─sda2        8:2    0    1K  0 part
└─sda5        8:5    0    1.1G  0 part [SWAP]

```

Exercise 12 – Explore Linux Log Storage

Investigate `/var/log` and identify:

- Available log files
- Available log directories
- Recently modified entries
- Entries you can read
- Entries you cannot read as a normal user

Commands and Meaning

1. List available log files and directories

```
ls -lah /var/log
```

Meaning: Lists all entries with detailed information and human-readable sizes.

2. Find log files

```
find /var/log -maxdepth 1 -type f
```

Meaning: Finds regular files directly inside `/var/log`.

(gopika@kali)-[~/linux/theory/t3]

\$ ls -lah /var/log

```
total 1.8M
drwxr-xr-x 22 root root 4.0K Sep  4 22:35 .
drwxr-xr-x 12 root root 4.0K Aug 11 01:16 ..
-rw-r--r--  1 root root 85K Aug 11 01:31 alternatives.log
drwxr-x--  2 root adm 4.0K Aug 11 01:19 apache2
drwxr-xr-x  2 root root 4.0K Aug 22 17:18 apt
-rw-----  1 root root 33K Sep  4 22:34 boot.log
-rw-----  1 root root 14K Aug 28 15:11 boot.log.1
-rw-----  1 root root 90K Aug 25 11:20 boot.log.2
-rw-rw----  1 root utmp 1.2K Aug 21 19:28 btmp
-rw-r--r--  1 root root 1.4M Aug 22 17:19 dpkg.log
-rw-r--r--  1 root root 7.2K Aug 11 01:26 fontconfig.log
drwxr-xr-x  2 _gvm _gvm 4.0K Apr 28 19:26 gvm
drwx-----  3 inetsim inetsim 4.0K Aug 11 01:20 inetsim
drwxr-xr-x  3 root root 4.0K Aug 11 01:33 installer
drwxr-sr-x+  3 root systemd-journal 4.0K Aug 11 01:15 journal
-rw-rw-r--  1 root utmp 0 Aug 11 00:47 lastlog
drwx--x--x  2 root root 4.0K Sep  4 22:34 lightdm
-rw-r--r--  1 root root 1.8K Sep  4 22:34 macchanger.log
drwxr-----  2 mosquitto mosquitto 4.0K May 14 15:13 mosquitto
drwxr-s--  2 mysql adm 4.0K Aug 11 01:17 mysql
drwxr-xr-x  2 root adm 4.0K Aug 11 01:10 nginx
drwxr-xr-x  2 _gvm _gvm 4.0K Apr 16 11:12 notus-scanner
drwxr-xr-x  2 root root 4.0K Apr 28 02:55 openvpn
drwxrwxr-t  2 root postgres 4.0K Aug 11 01:19 postgresql
drwx-----  2 root root 4.0K Aug 11 00:48 private
drwxr-s--  2 redis adm 4.0K Aug 11 01:12 redis
drwxr-xr-x  4 root root 4.0K Aug 11 01:09 runit
drwxr-x--  2 root adm 4.0K May 26 18:16 samba
drwx-----  2 speech-dispatcher root 4.0K Aug 11 01:19 speech-dispatcher
drwxr-xr-x  2 stunnel stunnel 4.0K Aug 11 01:13 stunnel
drwxr-xr-x  2 root root 4.0K Jan 16 2024 sysstat
-rw-rw-r--  1 root utmp 19K Sep  4 22:35 wtmp
-rw-r--r--  1 root root 8.0K Sep  4 22:35 wtmp.db
-rw-r--r--  1 root root 25K Sep  4 22:34 Xorg.0.log
-rw-r--r--  1 root root 25K Sep  3 09:29 Xorg.0.log.old
```

(gopika@kali)-[~/linux/theory/t3]

\$ find /var/log -maxdepth 1 -type f

```
/var/log/lastlog
/var/log/alternatives.log
/var/log/fontconfig.log
/var/log/wtmp.db
/var/log/Xorg.0.log
/var/log/boot.log.2
/var/log/macchanger.log
/var/log/boot.log
```

```

(gopika@kali)-[~/linux/theory/t3]
$ find /var/log -maxdepth 1 -type d | head -n 5
/var/log
/var/log/private
/var/log/inetd
/var/log/redis
/var/log/runit

(gopika@kali)-[~/linux/theory/t3]
$ ls -lt /var/log | head -n 5
total 1812
-rw-rw-r-- 1 root      utmp           18816 Sep  4 22:35 wtmp
-rw-rw-r-- 1 root      root            8192 Sep  4 22:35 wtmp.db
drwx-x-x 2 root      root            4096 Sep  4 22:34 lightdm
-rw-rw-r-- 1 root      root          25348 Sep  4 22:34 Xorg.0.log

(gopika@kali)-[~/linux/theory/t3]
$ find /var/log -maxdepth 1 -readable -print | head -n 5
/var/log
/var/log/lastlog
/var/log/alternatives.log
/var/log/fontconfig.log
/var/log/redis

(gopika@kali)-[~/linux/theory/t3]
$ find /var/log -maxdepth 1 ! -readable -print | head -n 5
/var/log/private
/var/log/inetd
/var/log/boot.log.2
/var/log/lightdm
/var/log/boot.log

```

3. Find log directories

```
find /var/log -maxdepth 1 -type d
```

Meaning: Finds directories directly inside `/var/log`.

4. Find recently modified entries

```
ls -lt /var/log
```

Meaning: Lists entries with the newest modified entries first.

5. Find readable entries

```
find /var/log -maxdepth 1 -readable -print
```

Meaning: Finds entries readable by the current user.

6. Find non-readable entries

```
find /var/log -maxdepth 1 ! -readable -print
```

Meaning: Finds entries that are not readable by the current user.

Meaning of Options

Option	Meaning	Option	Meaning
<code>-l</code>	Long/detailed listing	<code>-a</code>	Show all entries
<code>-h</code>	Human-readable sizes	<code>-maxdepth 1</code>	Search only specified directory
<code>-type f</code>	Find files	<code>-type d</code>	Find directories
<code>-t</code>	Newest modified first	<code>-readable</code>	Find readable entries
<code>!</code>	NOT condition	<code>-print</code>	Print matching entries

Why No `sudo`?

Do not use `sudo` when checking readable and non-readable entries because the exercise asks what a **normal user** can access. `sudo` gives administrator privileges and can change the result.

Answers

1. Why does Linux maintain logs?

Linux maintains logs to record system, application, and security events. These records help track what happened on the system.

2. Why are logs useful during incident investigation?

Logs provide evidence of activities such as logins, errors, and suspicious events. They help investigators reconstruct a timeline of an incident.

3. Why might access to some logs be restricted?

Some logs contain sensitive system or security information. Restricting access prevents unauthorized users from viewing or modifying them.

Exercise 13 – Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Determine:

- Whether files/directories currently exist.
- Whether hidden files exist.
- Who owns some of the entries.
- Which entries were modified recently.
-

Commands and Meaning

1. List files and directories

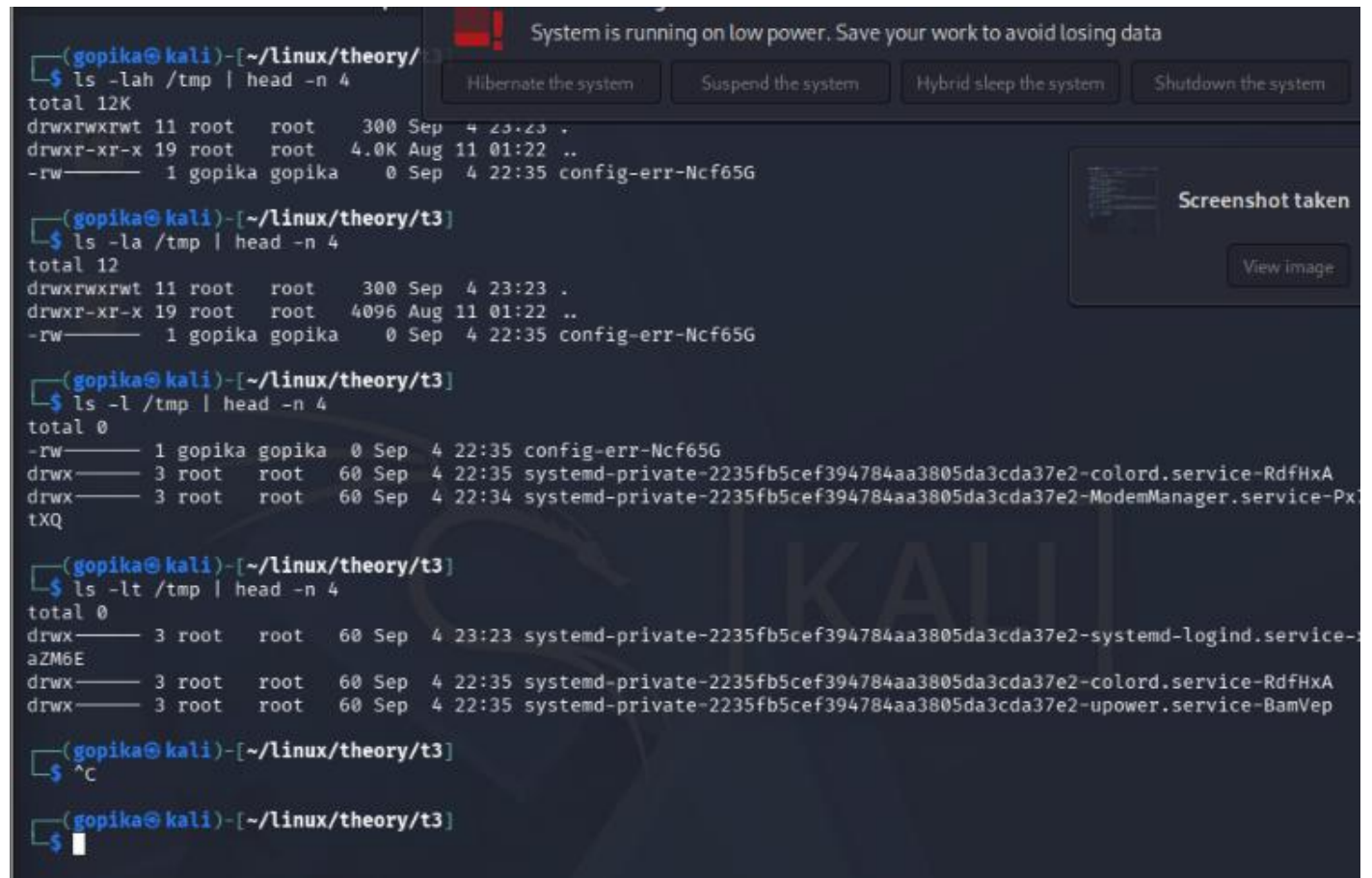
```
ls -lah /tmp | head -n 4
```

Meaning: Lists `/tmp` entries with details, hidden entries, and human-readable sizes.

2. Check hidden files

```
ls -la /tmp | head -n 4
```

Meaning: Lists all entries, including hidden entries.



```
(gopika@kali)-[~/linux/theory/t3]
$ ls -lah /tmp | head -n 4
total 12K
drwxrwxrwt 11 root root 300 Sep  4 23:23 .
drwxr-xr-x 19 root root 4.0K Aug 11 01:22 ..
-rw----- 1 gopika gopika  0 Sep  4 22:35 config-err-Ncf65G

(gopika@kali)-[~/linux/theory/t3]
$ ls -la /tmp | head -n 4
total 12
drwxrwxrwt 11 root root 300 Sep  4 23:23 .
drwxr-xr-x 19 root root 4096 Aug 11 01:22 ..
-rw----- 1 gopika gopika  0 Sep  4 22:35 config-err-Ncf65G

(gopika@kali)-[~/linux/theory/t3]
$ ls -l /tmp | head -n 4
total 0
-rw----- 1 gopika gopika  0 Sep  4 22:35 config-err-Ncf65G
drwx----- 3 root root 60 Sep  4 22:35 systemd-private-2235fb5cef394784aa3805da3cda37e2-color.service-RdfHxA
drwx----- 3 root root 60 Sep  4 22:34 systemd-private-2235fb5cef394784aa3805da3cda37e2-ModemManager.service-Px
tXQ

(gopika@kali)-[~/linux/theory/t3]
$ ls -lt /tmp | head -n 4
total 0
drwx----- 3 root root 60 Sep  4 23:23 systemd-private-2235fb5cef394784aa3805da3cda37e2-systemd-logind.service-
aZM6E
drwx----- 3 root root 60 Sep  4 22:35 systemd-private-2235fb5cef394784aa3805da3cda37e2-color.service-RdfHxA
drwx----- 3 root root 60 Sep  4 22:35 systemd-private-2235fb5cef394784aa3805da3cda37e2-upower.service-BamVep

(gopika@kali)-[~/linux/theory/t3]
$ ^C

(gopika@kali)-[~/linux/theory/t3]
$
```

3. Check ownership

```
ls -l /tmp | head -n 4
```

Meaning: Shows permissions, owner, group, size, and modification time.

4. Find recently modified entries

```
ls -lt /tmp | head -n 4
```

Meaning: Lists entries with the newest modified entries first.

Meaning of Options

Option	Meaning	Option	Meaning
-l	Long/detailed listing	-a	Show all entries
-h	Human-readable sizes	-t	Newest modified first
head -n 4	Show first 4 lines	/tmp	Directory being investigated

Observation Table

Investigation	Result from Your Output
Files/directories exist	Yes, files and directories are present
Hidden entries	. and .. are present
Ownership	config-err-Ncf65G → gopika:gopika; systemd-private-* → root:root
Recently modified	systemd-private-...systemd-logind... → 23:23, the newest shown

How to Identify Ownership

```
-rw----- 1 gopika gopika 0 Sep 4 22:35 config-err-Ncf65G
`gopika` = **owner**
second `gopika` = **group**

drwx----- 3 root root 60 Sep 4 23:23 systemd-private-...
`root` = **owner**
second `root` = **group**
```

Answer

Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories may contain temporary files, scripts, downloads, or other artifacts created during suspicious activity. They can provide useful evidence about recent system or user activity.

Exercise 14 – Explore the `/proc` Filesystem

Investigate: `/proc`

Examine:

```
cat /proc/cpuinfo
cat /proc/meminfo
```

Then identify your current shell PID:

```
echo $$
```

Check whether a corresponding directory exists under `/proc`.

```

(gopika@kali)-[~/linux/theory/t3]
$ cat /proc/cpuinfo | head -n 4
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 23
model          : 160

(gopika@kali)-[~/linux/theory/t3]
$ cat /proc/meminfo | head -n 5
MemTotal:      2016604 kB
MemFree:       299724 kB
MemAvailable:  1209664 kB
Buffers:       89740 kB
Cached:        919504 kB

(gopika@kali)-[~/linux/theory/t3]
$ echo $$
2030

(gopika@kali)-[~/linux/theory/t3]
$ ls -ld /proc/$$
dr-xr-xr-x 9 gopika gopika 0 Sep  4 22:39 /proc/2030

```

Commands and Meaning

Task	Command	Output / Observation	Meaning
Examine CPU information	cat /proc/cpuinfo head -n 4	processor : 0; vendor_id : AuthenticAMD; cpu family : 23; model : 160	Displays CPU information such as processor number, vendor, family, and model.
Examine memory information	cat /proc/meminfo head -n 5	MemTotal: 2016604 kB; MemFree: 299724 kB; MemAvailable: 1209664 kB; Buffers: 89740 kB; Cached: 919504 kB	Displays system memory statistics.
Identify current shell PID	echo \$\$	2030	2030 is the PID of the current shell.
Check the PID directory	ls -ld /proc/\$\$	dr-xr-xr-x 9 gopika gopika 0 Sep 4 22:39 /proc/2030	Confirms that a directory for the current shell process exists under `/proc`.

Answers

1. What information does `/proc/cpuinfo` provide?

It provides CPU details such as the processor number, vendor, CPU family, and model.

2. What information does `/proc/meminfo` provide?

It provides memory statistics such as total, free, available, buffered, and cached memory.

3. Why do directories with numbers appear inside `/proc`?

They represent running processes. Each number is a process ID (PID).

4. Is `/proc` an ordinary disk directory?

No. `/proc` is a virtual filesystem created by the Linux kernel.

5. Why could `/proc` be useful during security investigation?

It provides information about running processes and system resources, which can help identify suspicious activity.

Exercise 15 – Explore `/dev`

Investigate `/dev` and examine `/dev/null` and `/dev/zero`.

Commands and Meaning

1. Examine `/dev/null`

```
ls -l /dev/null
```

Meaning: Displays the permissions, owner, group, and type of `/dev/null`.

2. Examine `/dev/zero`

```
ls -l /dev/zero
```

Meaning: Displays the permissions, owner, group, and type of `/dev/zero`.

3. Determine the file type

```
file /dev/null
```

Meaning: Identifies the type of `/dev/null`.

Task	Command	Output / Observation	Meaning
Check `/dev`	/dev	bash: /dev: Is a directory	`/dev` is a directory and cannot be executed as a command.
Check `/dev/null`	/dev/null	bash: /dev/null: Permission denied	`/dev/null` is a special device file, not a normal executable file.
Examine `/dev/null`	ls -l /dev/null	crw-rw-rw- 1 root root 1, 3 Sep 4 22:34 /dev/null	First character `c` shows a character device.
Examine `/dev/zero`	ls -l /dev/zero	crw-rw-rw- 1 root root 1, 5 Sep 4 22:34 /dev/zero	First character `c` shows a character device.

Meaning of Options

Option	Meaning	Option	Meaning
-l	Long/detailed listing	/dev	Device filesystem
c	Character device	file	Identifies file type

Answers

1. What does `/dev` represent?

/dev contains special files representing hardware and virtual devices used by Linux.

2. Are the entries ordinary user files?

No. They are special device files used by the operating system to communicate with devices.

3. Determine the file type of `/dev/null`.

/dev/null is a **character device**, indicated by `c` at the beginning of the `ls -l` output.

4. Why does Linux expose devices through the filesystem?

It provides a standard way for programs and users to interact with hardware and virtual devices using filesystem operations.

```
(gopika@kali)~[/linux/theory/t3]
$ ls -l /dev | head -n 5
total 0
crw-r--r--  1 root root    10, 235 Sep  4 22:34 autofs
drwxr-xr-x  2 root root    140 Sep  4 23:23 block
drwxr-xr-x  2 root root     80 Sep  4 22:34 bsg
crw-----  1 root root    10, 234 Sep  4 22:34 btrfs-control

(gopika@kali)~[/linux/theory/t3]
$ ls -l /dev/null | head -n 4
crw-rw-rw-  1 root root 1, 3 Sep  4 22:34 /dev/null

(gopika@kali)~[/linux/theory/t3]
$ ls -l /dev/zero
crw-rw-rw-  1 root root 1, 5 Sep  4 22:34 /dev/zero

(gopika@kali)~[/linux/theory/t3]
$ file /dev/null
/dev/null: character special (1/3)

(gopika@kali)~[/linux/theory/t3]
$ file /dev/zero
/dev/zero: character special (1/5)

(gopika@kali)~[/linux/theory/t3]
$
```

Exercise 16 – Filesystem Space Investigation

Determine:

- Total filesystem capacity
- Used space
- Available space
- Percentage of filesystem utilization
- Space occupied by your home directory
- Space occupied by the `cyber\_lab` directory

Use appropriate combinations of:

```
df
df -h
du
du -h
du -sh
```

Commands and Meaning

Task	Command	Meaning
Check filesystem space	df	Shows filesystem capacity, used space, available space, and utilization.
Human-readable filesystem space	df -h	Displays filesystem sizes in readable units such as GB and MB.
Check directory space	du	Shows disk space used by files and directories.
Human-readable directory space	du -h	Displays directory usage in readable units.
Check total directory size	du -sh	Shows only the total size of the specified directory.

Commands for Required Information

1. Check filesystem space

```
df -h
```

Meaning: Shows total, used, available space, and usage percentage in human-readable format.

Use the row containing `/` to record the filesystem containing your root directory.

2. Check home directory space

```
du -sh "$HOME"
```

Meaning: Shows the total space occupied by your home directory.

3. Check `cyber\_lab` directory space

```
du -sh ~/linux/theory/t3/cyber_lab
```

Meaning: Shows the total space occupied by the `cyber\_lab` directory.

```
(gopika@ kali)-[~/linux/theory/t3]
$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
udev            972828         0   972828    0% /dev
tmpfs           201664        980   200684    1% /run
/dev/sda1      19353424 15597372  2747620   86% /
tmpfs          1008300         4  1008296    1% /dev/shm
tmpfs          1008304         8  1008296    1% /tmp
none            1024         0    1024    0% /run/credentials/getty@tty1.service
tmpfs          201660        108   201552    1% /run/user/1000
none            1024         0    1024    0% /run/credentials/systemd-journald.service

(gopika@ kali)-[~/linux/theory/t3]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            951M     0  951M   0% /dev
tmpfs           197M  980K  196M   1% /run
/dev/sda1       19G   15G  2.7G  86% /
tmpfs           985M   4.0K  985M   1% /dev/shm
tmpfs           985M   8.0K  985M   1% /tmp
none            1.0M     0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  108K  197M   1% /run/user/1000
none            1.0M     0   1.0M   0% /run/credentials/systemd-journald.service

(gopika@ kali)-[~/linux/theory/t3]
$ du
8   ./cyber_lab/reports
4   ./cyber_lab/evidence/network
4   ./cyber_lab/evidence/system
12  ./cyber_lab/evidence
4   ./cyber_lab/logs
4   ./cyber_lab/scripts
32  ./cyber_lab
36  .

(gopika@ kali)-[~/linux/theory/t3]
$ du -h
8.0K  ./cyber_lab/reports
4.0K  ./cyber_lab/evidence/network
4.0K  ./cyber_lab/evidence/system
12K   ./cyber_lab/evidence
4.0K  ./cyber_lab/logs
4.0K  ./cyber_lab/scripts
32K   ./cyber_lab
36K   .

(gopika@ kali)-[~/linux/theory/t3]
$ du -sh
36K   .
```

Observation

Item	Result
Total filesystem capacity	From `df -h`
Used space	From `df -h`
Available space	From `df -h`
Filesystem utilization	From `df -h`
Home directory size	From `du -sh "\$HOME"`
`cyber_lab` size	From `du -sh ~/linux/theory/t3/cyber_lab`

Answer

`df` shows space available and used on the **filesystem as a whole**, while `du` shows space used by **individual files and directories**.

```

(gopika@kali)~[/linux/theory/t3]
$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
udev            972828          0      972828   0% /dev
tmpfs           201664          980      200684   1% /run
/dev/sda1       19353424 15597372  2747620  86% /
tmpfs           1008300          4     1008296   1% /dev/shm
tmpfs           1008304          8     1008296   1% /tmp
none            1024            0       1024   0% /run/credentials/getty@tty1.service
tmpfs           201660         108     201552   1% /run/user/1000
none            1024            0       1024   0% /run/credentials/systemd-journald.service

(gopika@kali)~[/linux/theory/t3]
$ df -h
Filesystem      Size      Used Avail Use% Mounted on
udev            951M          0      951M   0% /dev
tmpfs           197M        980K      196M   1% /run
/dev/sda1       19G         15G     2.7G  86% /
tmpfs           985M        4.0K      985M   1% /dev/shm
tmpfs           985M        8.0K      985M   1% /tmp
none            1.0M          0       1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M       108K      197M   1% /run/user/1000
none            1.0M          0       1.0M   0% /run/credentials/systemd-journald.service

(gopika@kali)~[/linux/theory/t3]
$ du
8      ./cyber_lab/reports
4      ./cyber_lab/evidence/network
4      ./cyber_lab/evidence/system
12     ./cyber_lab/evidence
4      ./cyber_lab/logs
4      ./cyber_lab/scripts
32     ./cyber_lab
36     .

(gopika@kali)~[/linux/theory/t3]
$ du -h
8.0K    ./cyber_lab/reports
4.0K    ./cyber_lab/evidence/network
4.0K    ./cyber_lab/evidence/system
12K     ./cyber_lab/evidence
4.0K    ./cyber_lab/logs
4.0K    ./cyber_lab/scripts
32K     ./cyber_lab
36K     .

(gopika@kali)~[/linux/theory/t3]
$ du -sh
36K     .

```

```

(gopika@kali)~[/linux/theory/t3]
$ du -sh "$HOME"
81M     /home/gopika

(gopika@kali)~[/linux/theory/t3]
$ du -sh ~/linux/theory/t3/cyber_lab
32K     /home/gopika/linux/theory/t3/cyber_lab

```

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported. Investigate the following filesystem locations and identify why each is important.

Investigation Table

Location	Why Investigate?	Security-Relevant Information
/home	Contains files and directories belonging to normal users.	Suspicious files, scripts, downloads, hidden files, and user activity.

Location	Why Investigate?	Security-Relevant Information
/etc	Contains important system and service configuration files.	User accounts, authentication, networking, services, and system configuration.
/var/log	Contains system and application logs.	Login activity, errors, service events, and suspicious activity.
/tmp	Contains temporary files created by users and applications.	Temporary scripts, downloads, files, and possible attack artifacts.
/proc	Provides information about the running system and processes.	Running processes, PIDs, memory information, and current system activity.
/dev	Contains special files representing devices.	Device information and unusual or suspicious device entries.
/root	Contains files belonging to the root user.	Root-user files, scripts, configuration files, and possible suspicious activity.

Conclusion

All these locations are relevant during a filesystem-based cybersecurity investigation. Together, they can provide evidence about ****users, configurations, logs, temporary artifacts, running processes, devices, and root-user activity****.

Submit the following

- Commands used for each exercise.
- Relevant output or screenshots.
- Observation tables.
- Disk → Partition → Filesystem → Mount Point diagram.
- Answers to analysis questions.

****Important:**** Do not modify or delete system files while performing the exercises.

Linux Filesystem – Command Cheat Sheet

Exercise	Commands
1. Filesystem	lsblk → devices/partitions lsblk -f → filesystem type df -T → filesystem & space df -h → readable space
2. Storage Diagram	Use output from Exercise 1 → Disk → Partition → Filesystem → Mount Point
3. Root Filesystem	pwd → current path cd / → root ls → contents ls -la → all entries
4. Hierarchy	ls -lah /home ls -lah /root ls -lah /etc ls -lah /var ls -lah /var/log ls -lah /tmp ls -lah /usr ls -lah /boot ls -lah /dev ls -lah /proc
5. Home Directory	whoami → username echo \$HOME → home path cd "\$HOME" → home pwd → absolute path ls → normal contents ls -a → all contents
6. Hidden Files	ls → normal entries ls -a → hidden entries
7. Build Hierarchy	mkdir -p cyber_lab/evidence/network cyber_lab/evidence/system cyber_lab/logs cyber_lab/scripts cyber_lab/reports touch cyber_lab/evidence/network/traffic.txt touch cyber_lab/evidence/system/processes.txt touch cyber_lab/logs/security.log touch cyber_lab/scripts/check.sh touch cyber_lab/reports/investigation.txt tree cyber_lab
8. Paths	cd evidence/network → relative cd /etc → absolute cd ~ → home cd ~/cyber_lab/reports cd .. → one level up pwd → current path
9. File Types	file <filename> → type ls -l <filename> → details
10. Metadata	echo "Investigation report" > investigation.txt ls -l investigation.txt stat investigation.txt
11. /etc	ls -lah /etc cat /etc/passwd cat /etc/group cat /etc/hostname cat /etc/hosts
12. /var/log	ls -lah /var/log find /var/log -maxdepth 1 -type f find /var/log -maxdepth 1 -type d ls -lt /var/log find /var/log -maxdepth 1 -readable -print find /var/log -maxdepth 1 ! -readable -print
13. /tmp	ls -lah /tmp ls -la /tmp ls -l /tmp ls -lt /tmp
14. /proc	cat /proc/cpuinfo head -n 4 cat /proc/meminfo head -n 5 echo \$\$ ls -ld /proc/\$\$

Exercise	Commands
15. /dev	ls -l /dev/null ls -l /dev/zero file /dev/null
16. Space	df df -h du du -h du -sh "\$HOME" du -sh ~/linux/theory/t3/cyber_lab
17. Investigation	ls -lah /home ls -lah /etc ls -lah /var/log ls -lah /tmp ls -lah /proc ls -lah /dev ls -lah /root

★ Quick Option Meanings

Option	Meaning	Option	Meaning
-l	Detailed listing	-a	All entries
-h	Human-readable	-t	Newest first
-f	Filesystem information	-T	Filesystem type
-s	Summarize size	-d	Directory itself
-p	Create parent directories	-maxdepth 1	Only specified directory
-type f	Files	-type d	Directories
-readable	Readable entries	!	NOT condition
-print	Print results	\$\$	Current shell PID